

Three objects, and what they were found to do

Findings on the sound produced by the recovered artefacts B2311.22, B2311.67 and B2311.104: what has been established, what is suspected, and what remains entirely unclear.

Primary	Recovered	Sites	Status
M5.5Ve · 0.0485 AU	3 of 3 reported	Kell Rille · Sabik Terminator	All objects active

- ESTABLISHED

Measured, repeated, and quotable. Every figure marked this way came off an instrument, and the measurement has been made again.
- SUSPECTED

Consistent with everything measured and not demonstrated. Marked as such wherever it appears, including where it is the most interesting thing on the page.
- UNRESOLVED

The survey could not close it. Some of these we know how to ask; at least one we do not.

ONE · THE SITUATION

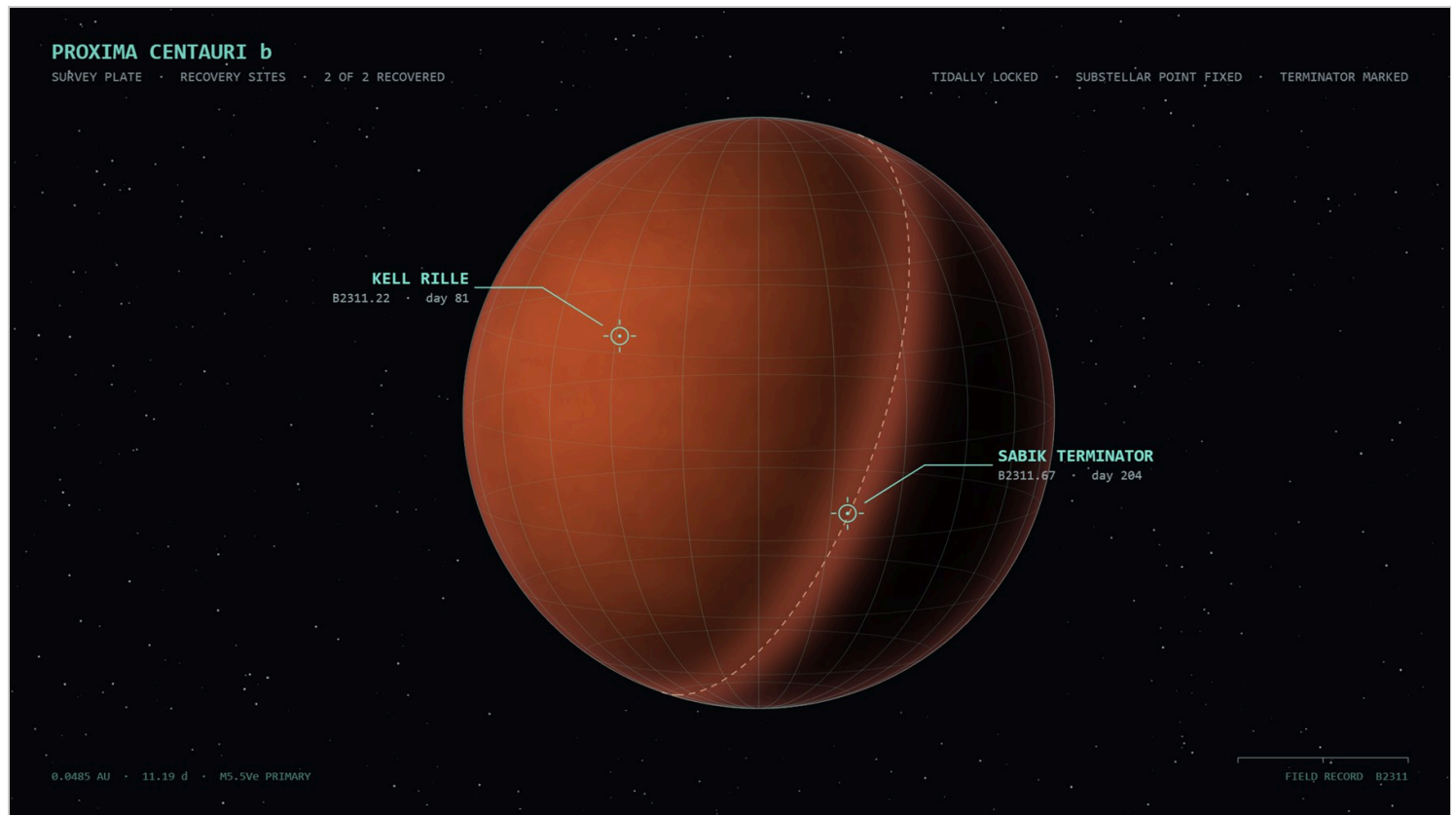
What was found, and in what condition

Neither object was made by us, and neither resembles anything we make. Both were recovered intact, both were sounding when recovered, and neither has been persuaded to stop.

B2311.22 was lifted on day 81 from a dry cistern at Kell Rille, on the lit side. It did not come up alone. A second body lay beside it, and on raising the first, the second changed pitch — an interaction recorded before either had been moved more than a metre, and the first indication that these objects are aware of one another.

B2311.67 was recovered 123 days later from a vault field at the Sabik Terminator, in the permanent twilight where the day side meets the night. There were sixty-seven of them. Each sat at the centre of its own void, spaced with a regularity that

survives measurement, and no two voids touched.



Survey plate. The world is tidally locked: the substellar point never moves and the terminator is a permanent ring rather than a passing hour. Kell Rille lies well inside the lit hemisphere; the Sabik vault field lies on the boundary itself.

The first thing established about both objects, and the thing that has shaped every attempt to work with them since, is that *what we are holding is not the whole object*. Each is a body of four spatial dimensions. What the hand meets is a three-dimensional section through it, and what the eye is shown is a two-dimensional projection of that section. Move the section and the object is not altered — a different part of it is simply present. Tissue and structure enter and leave existence at the boundary, continuously, without damage and without any sign that the object regards this as an event.

The second thing established is that *neither object has controls*. There are no markings, no surfaces that invite a hand, nothing that moves when pushed. Everything named in this report — every quantity, every reading, every word on the frames the expedition built around them — is ours. The objects present nothing. The apparatus exists so that a human being can act on them at all, and it should not be mistaken for part of them.

An object whose sound is its own shape

The sound of B2311.22 is the vibration of its own connectedness. It is not struck like a bell or blown like a pipe; it is a network, and what is heard is that network ringing according to how its parts are joined. Change which point is disturbed and the sound changes, because a different part of the structure is being asked to move.

Established. Disturbing the object at its most connected point, versus its most remote, produces spectra differing by 0.51 in relative terms – the same object, two different voices, depending only on where it is touched.

The frequencies it produces do not form any series we use. There is no octave in them and no harmonic ladder; the intervals are those of the object's own construction and nothing else. This was not assumed but measured across the whole catalogue, and it holds without exception: *no specimen sits within twenty cents of any harmonic series*. The closest approach found anywhere in the collection lands exactly on that floor, and no nearer.



B2311.22, specimen 037. Bright tissue is inside the section and sounding; the two dim forms at centre are structure approaching the boundary but not yet present, and they will brighten without anything having been done to the object. The brightness of a region is that region's share of the energy – so this plate is, in part, a picture of what is being heard. Only in part: see section seven.

Energy does not stay where it is put

The behaviour that most clearly distinguishes this object from anything of ours is what happens after it is disturbed. Our instruments decay: a struck string loses energy and falls silent. B2311.22 does not lose the energy. It *moves* it, along its own structure, from one mode of vibration into its neighbours, so that a sustained note is not a note being held but the object exploring itself.

Established. Over a single sustained disturbance the centre of the sound's energy travels from 645 Hz to 582 Hz without any further input. The object is not decaying. It is redistributing.

Exact recurrence

Under one configuration of the frame the object's output becomes exactly periodic — not approximately, but exactly. Every detail of the sound reassembles itself after an interval between one and a half and four seconds, depending on the specimen, and the reassembly is faithful to a correlation of 0.997. In the interval between recurrences, fractional partial states are audible that belong to neither the beginning nor the end.

The second body

The pair-recovery at Kell Rille was not an accident of storage. When a second object is placed within the same vessel it *responds*, and the character of the response is the most important finding of the survey.

It does not hear everything. It absorbs only at the frequencies it itself possesses, so how well two objects can hear one another is a fixed property of that pair. Two closely related bodies converse fluently; two unrelated ones are very nearly deaf to each other. When it answers, it answers *in its own frequencies* — the shape of what it heard, re-spoken through a different anatomy. The same statement, in another body's voice.

Established. Ladder overlap between an object and itself measures 1.000; between the least related pair in the catalogue, 0.180. Closely related bodies answer 2.2 times more strongly than unrelated ones — but no pair is ever mute. A darker utterance draws a reply centred at 386 Hz; a brighter one, 406 Hz, from the same responding body.

While both are sounding together, a third set of frequencies appears which belongs to neither of them and cannot be produced by either alone. It exists only for as long as both are sounding, is different for every pair, and vanishes the instant one falls silent. It accounts for roughly a sixth of what is heard during an exchange.

The finding we cannot account for

Being spoken to changes the listener. Not its behaviour — *its structure*. After a sustained exchange the responding body's own frequencies have moved, and they do not move back. The change accumulates in the direction of what it heard.

Established. Thirty seconds of exchange displaced a responding body's own frequencies by 53.9 cents, in nineteen discrete steps, each applied while the object was at rest. The displacement is permanent. An object that has been in a vessel with another for long enough no longer matches its own catalogue entry.

We have no account of the mechanism and no way to reverse it. It should be assumed that every object recovered has already been altered by whatever it spent time beside.

THREE · B2311.67

An object that is ordered and never repeats

The Sabik object is not a body in the sense that B2311.22 is a body. It is a lattice: a regular four-dimensional array, of which we again meet only a section. The section is a tiling of squares and rhombi with eightfold symmetry — perfectly ordered, and never repeating anywhere in its extent.

This is the more difficult of the two objects to describe, because the category its sound belongs to has no name in ordinary use. Sound as we make it is of two kinds. There are tones, in which energy sits at separated frequencies — every string, pipe, bell and oscillator ever built. And there is noise, in which energy is spread continuously — breath, hiss, the sea. Mathematics has always described a third possibility between them, in which energy occupies a set of frequencies that is infinitely subdivided and yet takes up no width at all: a dust with self-similar gaps at every scale.

No material we have ever made produces it. This object does.

Established. Energy injected at one point of the lattice spreads neither as heat spreads nor as a wave travels, but by a power law between the two. The sound arrives smeared in a manner that no room, no enclosure and no material in our possession reproduces.

The lengths at which the structure repeats its own logic are not arbitrary. They follow the sequence 5, 12, 29, 70, 169, 408 — the ratios native to eightfold order, as inevitable to this object as the octave is absent from it.

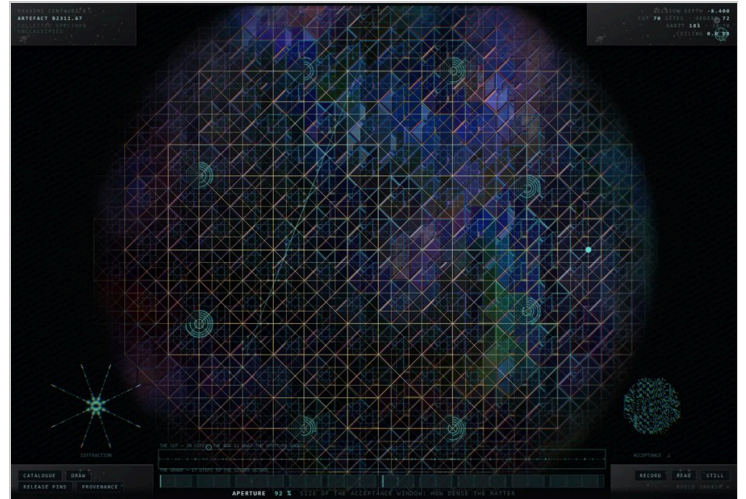
Two hundred and fifty-six specimens, and one body

The catalogue holds 256 specimens, and they sound so unlike one another that they were at first taken to be different structures. They are not. The survey measured the geometry of every one of them and found no variation whatever.

Established. The proportion of squares to rhombi is 0.438 in every specimen in the catalogue — smallest, median and largest identical across all 256. What distinguishes one

specimen from another is never *which* structure exists, only how that structure is weighted. Every specimen is the same body wearing different material.

The figure the object casts changes with how much of the lattice is admitted, and this is the most legible thing on the frame: opened narrowly it throws a four-armed cross, opened wide, a full eightfold star. The temptation is to call this a change of symmetry, and the survey record must not, because the measurement says otherwise.



The same specimen at 12 per cent and 92 per cent of the acceptance window; the figure it casts is the small form at the lower left of each plate. By bond angle the tiling is exactly eightfold in both — measured at every depth, every window and every specimen without exception. What actually changes is the mixture: almost entirely square-dominated when narrow, roughly a third so when wide. The figure follows the mixture, not the symmetry, and this report therefore does not describe it as a diffraction pattern.

The strain that makes a crystal

One quantity on the frame does something none of the others do. Where travelling through the fourth dimension exchanges one arrangement for another equally aperiodic, this one *tilts the section against the lattice* — and a tilted cut through an ordered aperiodic body is the classical route by which such a body becomes an ordinary crystal. The object goes willingly.

Established. Tilting the section moves the sustained sound roughly twenty times as far as travelling through the fourth dimension does. Twenty of twenty-four specimens examined pass measurably nearer a harmonic series on the way, and one arrives within 0.2 cents of one — an object that had no octave in it becoming, briefly, an ordinary crystal with an ordinary voice.

This deserves to be set beside the corresponding finding for the other object. B2311.22 refuses the harmonic series absolutely: no specimen anywhere in its catalogue comes within twenty cents of one. B2311.67 will not produce a harmonic series either, until it is strained — and then it produces one to within a fifth of a cent.

Suspected. That the crystal is a destination rather than an accident. It lies at a particular tilt, it is reached by a smooth path, and the object arrives at it in tune. A structure that has one exact configuration in which it speaks our kind of interval, and no other, is either a coincidence of geometry or a door. The survey cannot tell which.

What is seen is not what is heard

The visible structure of B2311.67 and its sustained sound are two different voices, and they belong to different parts of the object. What the eye is given is struck, rings and dies; what the ear is given persists. For as long as a sound is held, the observer is watching one thing and listening to another.

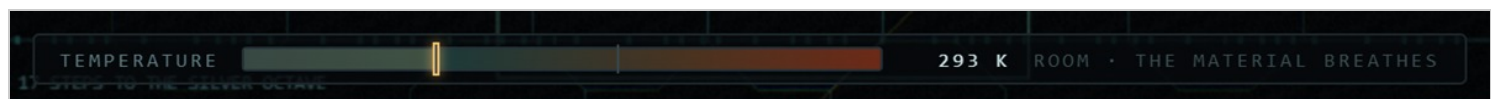
Established. With the reading drawn entirely from the visible structure, a held sound sustains at three orders of magnitude below the same sound drawn from the object's standing figure, and the visible array falls by a factor of sixteen thousand within three seconds. One quantity on the frame puts the visible body back into the held sound; for roughly three specimens in five it sits at zero unless a hand moves it.

FOUR • TEMPERATURE

The one condition that changes everything

Warming B2311.67 does not make it louder, brighter or faster. It makes the object's quantities begin to act on one another.

Held cold, every quantity of the object is independent: set it, and it stays where it was set, and nothing else moves. Warm the body and this stops being true. Quantities begin to drive other quantities — some in depth, some in rate — along connections that are fixed, that differ from specimen to specimen, and that no amount of handling will rearrange. The survey has recorded this as cross-talk for want of a better word, and section six gives the reason that word is probably wrong.



The only reading on the frame that is not a ring. The range runs from liquid nitrogen to a temperature at which no expedition material would survive; the object is indifferent to both ends.

TEMPERATURE	EFFECT ON THE MATERIAL	EFFECT ON THE FRAME OF OBSERVATION
77 K	none whatever	none
293 K	±8.96 % of range	none
500 K	±19 %	begins
800 K	±30 %	±12 %

Established. At 77 K the coupling is not merely small – it is absent, and the object is indistinguishable from one in which no such mechanism exists. Between there and room temperature the material stirs and nothing else does. Only above about 500 K do the quantities that frame the observation itself – where the section is taken, how it is angled, how much of the lattice is admitted – begin to drift on their own.

Established. Every specimen carries its own arrangement: four of its quantities are connected outward, each to two or three others. All 256 arrangements differ, and each is fixed to its specimen. The motion is slow – between one cycle in fifty seconds and one in two – a body settling rather than anything that could be mistaken for a tremor.

Established. The coupling is strictly one-way. No connection has ever been observed to read back what it wrote, and no closed loop has been found in any of the 256 arrangements. Whatever this mechanism is for, it is not for regulating itself.

Two further observations are worth recording precisely because they are negative. First, some quantities are never driven at any temperature the frame can reach: the object's loudness, its reference pitch, the identity of the specimen itself, and the size of the body. The object has never been observed to reach for these. Second, the response to heat is not uniform across the catalogue — at the top of the range 141 of the 256 specimens stir the frame of observation and the remaining 115 hold it perfectly still. That division is fixed per specimen and the survey cannot predict it from anything else it can measure.

FIVE • THE FRAME

What our apparatus can do, briefly

What follows is the shortest possible account of the expedition apparatus. It is short on purpose: the frame is ours, it is not interesting, and the rest of this report is about what it has not been able to settle.

Both objects are moved through the fourth dimension by a single continuous control. Nothing is created or destroyed by this; a different part of the object is brought into the section, and structure enters and leaves at the boundary as it does so.

B2311.22 answers the hand directly. Three kinds of contact have been distinguished and they produce different results: a circling motion, which accumulates and in which direction matters; a fast straight one, which disturbs and is then relaxed; and a still press held in one place, which deepens for as long as it is held and part of which remains after release. A separate control draws the section back towards where it was.

B2311.67 presents forty rings, arranged as eight rosettes of five, with no numerals, no pointers and no marked position. A ring is wound by dragging across it — right or up to raise, left or down to lower — with full travel in about a hand's width, and roughly four times finer with a modifier held. The rings sit on the body, so they travel with it, but they do not change places. The sounding line through the lattice can be dragged directly, a vertex can be arrested by touching it, and the temperature reading runs along the foot.

Note. Twenty-three of the forty-two quantities on the B2311.67 frame do not affect a sound already in progress. Most of these are honestly properties of a disturbance rather than of the object, and take effect at the next one. The remainder belong to the visible structure, which has already fallen silent by the time they are moved — see section three.

SIX • THE SUSPICION

What the couplings may be for

Two findings, made on two different objects by two different methods, point the same way, and the survey records the resemblance without being able to prove anything from it.

In B2311.22, an object placed beside another *answers* it — hearing only at the frequencies it itself possesses, replying in its own voice, and being permanently altered by the exchange. In B2311.67, warming the body makes its quantities drive one another along an arrangement that is fixed, unique to each specimen, one-way, and slow.

Suspected. That what the survey has recorded as cross-talk is not a defect of a warm material but a channel — that each specimen's arrangement of connections is its identity in the way a voice is an identity, and that temperature is simply the condition under which it speaks. On this reading the quantities on our frame are not settings at all. They are an alphabet, and the survey has spent two hundred days spelling at random.

Suspected. That the two objects are one technology in two conditions: one kept where it could converse, and sixty-seven kept where they could not. The vault spacing is exactly

what would be required to prevent the permanent alteration described in section two, and it is too regular to be incidental.

The survey takes both of these seriously enough to say what follows from them, which is a change in method. If these couplings are a channel, then nothing about them will be found by setting a quantity to a value and leaving it there, any more than a language is found by holding one note. Every finding in this report that turned out to matter was found by moving something and listening to what moved with it.

Whoever now holds these objects is therefore asked to explore rather than to configure — and to treat the search for couplings as the work itself, not as preparation for it.

In practice that means: hold one quantity and listen for what follows it, and for how long afterwards. Warm the body and note which quantities begin to move together, since that grouping is fixed and is different for every specimen. Return to the same specimen after time has passed, because at least one of these objects does not stay as it was found. Place two of them in one vessel and let them run without intervening. And attend particularly to what happens at the two thresholds the survey has located but cannot explain — the temperature at which the frame of observation itself begins to drift, and the tilt at which an aperiodic body becomes a crystal and speaks, for a moment, in our intervals.

SEVEN • UNRESOLVED

What the survey could not close

Question 1

Why 500 K? Below it the material stirs and the frame of observation is untouched; above it the frame drifts too. Nothing about the structure predicts a threshold there, and nothing predicts which specimens have one: 141 of 256 stir the frame at the top of the range and 115 hold it still, the division is fixed, and it does not correlate with any other property the survey can measure.

Question 2

What have the objects on the surface to do with the sound, and with one another? In B2311.67 what is seen and what is heard are demonstrably different voices. In B2311.22 the visible tissue does brighten where the energy sits — that much is established — but whether the visible features act on one another or merely display something decided elsewhere is not. The survey has no account of the interaction between the surface bodies at all: not with the sound, and not among themselves.

Question 3

Why does a listening body change permanently, and is the change legible? The displacement accumulates in the direction of what was heard. If it is a record rather than mere wear, then every recovered object carries the history of whatever stood beside it, and we do not know how to read it.

Question 4

Two objects were found together and interacting. Sixty-seven were found isolated, each at the centre of a deliberate void. Are these two conventions of storage, or two kinds of thing? The vault spacing is too regular to be incidental, and separation is exactly what would prevent the alteration described above.

Question 5

One specimen is not understood. The first habit in the B2311.67 catalogue drives its own resonance to a level some six hundred times what any other specimen produces, and the frame must hold it back by sixty decibels for it to be worked with at all. The restraint is reliable and has been verified as reliable. The behaviour has not been explained, and that specimen should not be treated as representative of anything.

Question 6

Neither object has an off state. Both were sounding when found, both have sounded continuously since, and the frame can only open onto that sound or close again. What the objects are doing when nothing is listening is not established, and the question may not be well posed.

Question 7

Is B2311.67 a specimen of ordered aperiodic matter, or a device that produces the behaviour of such matter? The distinction matters. The first would make it a sample; the second would make it an instrument, and would mean someone built it to sound like something.

Question 8

We can slide the section along one axis. We cannot rotate it. Everything reported here is what lies in one particular family of cross-sections through each object, and there is no reason to believe it is the informative one.

Question 9

We have found no controls, and have built our own. It has not been established whether the absence of controls is a fact about these objects or a fact about us — whether we are failing to find an interface, or whether the question of an interface does not arise for something that was never meant to be operated.

An object whose sound may be transported power

The most recent finding shares almost nothing with the other three, which is itself the most confounding thing about it. It is not a body and not a lattice but a *web*: slender conduits meeting at junctions, lying on the same curved four-dimensional surface, of which we meet a three-

dimensional section we can slide but not rotate. And it does not answer being struck. Left to itself it makes sound — deep, continuous, and its own.

The sound is generated, not excited

Established. A conduit of B2311.104 is silent until the disequilibrium across it crosses a threshold. Below the threshold, disturbed, it rings briefly and falls quiet. Above it, it begins to sound from nothing, grows to a steady amplitude, and holds there — the signature of a structure converting a standing gradient into acoustic power, not of a resonance being struck. Warmed past onset with nothing played into it, distant parts of the web begin to carry on their own. No other object in the record makes sound unbidden.

Heat is pitch

Established. The pitch of a conduit follows the square root of its temperature, across a fourfold change of the ambient, to within a few cents; a conduit held under command stays at the pitch it was given while everything around it retunes. Commanding a note, therefore, is commanding a temperature: the object heats a conduit until its resonance is the note asked for, at a rate we can set, through an inertia we can set, and cools it when the note is released. What we would call attack, portamento and release are, in this object, one thing — the warming and cooling of metal — and behave exactly as a thermal quantity should. Its voices lie between roughly 24 and 260 hertz; it is the deepest of the four.

The web is coupled, and the coupling carries

Established. Heat travels between conduits that share a junction, and sound leaks across the junctions as well. Playing one conduit hard measurably warms its neighbours, retunes them, and can carry them over their own threshold until they sound in sympathy — so a single note propagates through the structure as both heat and vibration along the very same connections.

What it may be for

Suspected. The survey's working hypothesis is that B2311.104 is a distribution device — that its builders moved power along these conduits as low-frequency sound, the way we move it as alternating current, and that what we hear is a fragment of their grid, still carrying. Every measurement is consistent with this. None demonstrates it. The survey records, without resolving, that no power apparatus it can conceive of would also be this beautiful to listen to, and that the difficulty may lie in the assumption that for its builders the two were different things.

On the frame, and on reading this report

Every quantity named in this report is an expedition quantity. The names on the frames were chosen by us, for our convenience, and describe what our apparatus does to the objects rather than any faculty the objects possess. Where this report says an object "answers" or "explores itself", these are the plainest available descriptions of measured behaviour and are not offered as claims about intent.

The three registers used here should be respected in any use made of this document. The temperature figures, the geometry, the spectral displacements and the permanence of the alteration are established and may be quoted. That the couplings constitute a channel is suspected, and is the reason the survey recommends the method it does; it is not a finding. The nine questions in section seven are open, and the second of them — what the visible bodies have to do with the sound, or with each other — is the one the survey has made least progress on and considers most likely to matter.

What can be said without qualification is that all three objects produce sound continuously, that the sound is lawful and repeatable, that it is generated by structures which have no counterpart among our instruments, that in one case the object is permanently altered by having been listened to and by having listened, and that in the newest the sound may not be a by-product of the object's working but the working itself.

Field record B2311 · Proxima Centauri b · findings compiled from the recovery of B2311.22 (Kell Rille, day 81), B2311.67 (Sabik Terminator, day 204) and B2311.104 (Kell Rille lower gallery, day 327).
All objects remain active. None has been disassembled.